

JUSTIN GU

✉ gujustin1@gmail.com | 🌐 quaot.github.io | 🐙 github.com/Quaot

EDUCATION

University of Waterloo

Bachelor of Applied Science, Honours Electrical Engineering

Waterloo, ON

Class of 2031

International Baccalaureate Diploma

Score: 40/45

Langley, BC

2026

EXPERIENCE

Project Lead, Bakfiets

Electrium Mobility

Sept. 2026 – Present

Waterloo, ON

- Lead the 25-person team finishing an electric cargo bike begun by the 2024 team, in a 100-member design club
- Merged the 2024–25 CAD, firmware and wiring files into one repo; wrote an onboarding guide and 30 GitHub issues
- Drew the planning diagrams for the bike's power flow, 12S3P battery pack and motor controller wiring

Academic Representative

Waterloo Engineering Society

Sept. 2026 – Present

Waterloo, ON

- Elected by a ~150-student ECE cohort as 1 of 6 academic reps; raise class issues with profs and the First Year Office

UI/UX Designer

ForgeX Trading

Jun. 2026 – Sept. 2026

Langley, BC

- Designed and shipped the live landing page for a trading replay platform, taking it from Figma to production
- Rebuilt result screens around one sortable table, cutting the clicks to compare two back-test runs from 6 to 2

Junior Robotics Coach

Zebra Robotics

Jun. 2023 – Jan. 2025

Langley, BC

- Coached students in robotics classes run in Scratch and Python over 19 months

PROJECTS

3.3V LDO Regulator PCB | Altium Designer (GitHub)

2026

- Stacked 1% resistor tolerance, reference spread and ADJ-pin current into a 3.20–3.44V worst case, inside $3.3V \pm 5\%$
- Chose a $15\mu F$ tantalum for the LM1117's $0.3\text{--}22\Omega$ output ESR window; derated the $22\mu F$ input ceramic to $\sim 10\mu F$ at 5V
- Budgeted 0.85W at 500mA; at $84^\circ C/W$ the calculated junction is $116^\circ C$ at $45^\circ C$ ambient, under the $125^\circ C$ recommended max

4-bit R-2R DAC | KiCad, SPICE, Oscilloscope (GitHub)

2026

- Drew a 4-bit R-2R ladder in KiCad, simulated all 16 codes in SPICE, checked each 206mV step on a scope in the E2 lab

3D Printing | Fusion 360, Cura, PLA (GitHub)

2025–2026

- Refined an IB Physics sensor mount over 5 insert versions, printing fit-test strips before the final 100mm holder

AI-Phillic – 1st of 20 Teams, IDEA-X \$10,000 Prize | React Native, Expo, TypeScript (GitHub)

2026

- Owned the app for a team of four (teammates ran the pitch): illustrated lessons that teach students to check sources

IB Paper Organizer | JavaScript, Electron (GitHub)

2024–2026

- Maintain a desktop app that sorts several hundred IB past papers by filename and pairs each with its mark scheme

TECHNICAL SKILLS

Hardware: Altium Designer, KiCad, Schematic Capture, 2-Layer PCB Layout, SPICE, Oscilloscope, Fusion 360, 3D Printing

Analysis: Thermal Budgeting, Datasheet Review, Capacitor Derating

Software: Python, TypeScript, JavaScript, HTML/CSS, React, React Native, Electron

Tools: Git, GitHub, Figma, Supabase